T6 — Organisms Respond to Changes in Their Environment

6.1 Stimuli and Receptors

Stimulus: A detectable change in the internal or external environment of an organism that produces a response.
Receptor: A cell, tissue, or organ that detects a stimulus and converts it into an electrical impulse (a **transducer** — converts one form of energy into another, specifically into nerve impulses).
Effector: A cell, tissue, or organ that carries out a response (muscles contract, glands secrete).

Organisms must detect and respond to changes in their environment to survive. The general pathway is:

$$\text{Stimulus} \rightarrow \text{Receptor} \rightarrow \text{Coordinator (CNS)} \rightarrow \text{Effector} \rightarrow \text{Response}$$

Types of Receptor

Receptor Type	Stimulus Detected	Examples
Photoreceptors	Light	Rods (dim light, black/white) and cones (colour, bright light) in the retina
Chemoreceptors	Chemicals	Taste buds (tongue), olfactory receptors (nasal epithelium), carotid body (blood pH/CO_2)
Mechanoreceptors	Pressure / movement / vibration	Pacinian corpuscles (deep pressure), hair cells in cochlea (sound), muscle spindles (stretch)

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Thermoreceptors	Temperature changes	Peripheral thermoreceptors in skin (warm and cold receptors), central thermoreceptors in hypothalamus (blood temperature)
Osmoreceptors	Water potential changes	Hypothalamus (detect changes in blood water potential/osmolarity)
Nociceptors	Pain / tissue damage	Free nerve endings throughout body

Pacinian Corpuscle — A Model Receptor (Transducer)

The Pacinian corpuscle is a pressure receptor found deep in the skin, joints, ligaments, and mesentery. It illustrates the general principle of how receptors generate a nerve impulse (the **generator potential**):

1. Pressure is applied, deforming the layered **lamellar** (onion-like) capsule surrounding the sensory neurone ending.
2. Deformation stretches the membrane of the sensory neurone ending (the node of Ranvier is exposed).
3. **Stretch-mediated (mechanically gated) sodium ion channels** in the neurone membrane open.
4. Na^+ ions flood into the neurone (down their electrochemical gradient), causing **depolarisation** of the membrane — this is the **generator potential**.
5. If the generator potential reaches the **threshold** (about -55 mV), voltage-gated Na^+ channels also open and an **action potential** (nerve impulse) is generated and transmitted along the sensory neurone to the CNS.
6. If the stimulus is too weak, the generator potential does not reach threshold, and no action potential is produced — this is a **subthreshold stimulus**.

Exam Tip: The Pacinian corpuscle demonstrates three key receptor properties: (1) it is a **transducer** — converts mechanical energy (pressure) into electrical energy (nerve impulse); (2) it shows **specificity** — only responds to pressure, not light or chemicals; (3) it demonstrates the **generator potential** mechanism that is common to all receptors.

The Eye — Photoreceptors

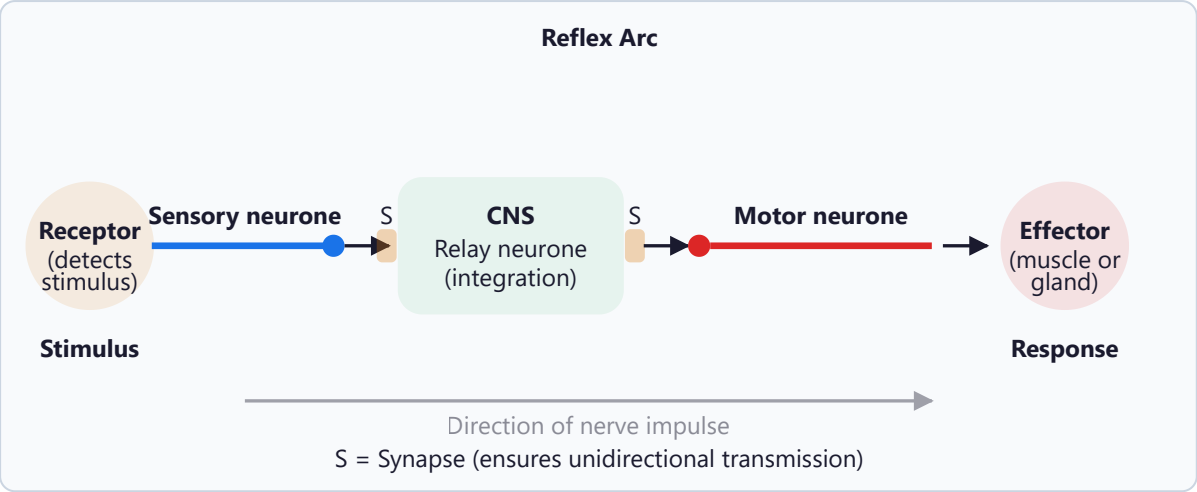
Light is focused by the cornea and lens onto the retina, which contains two types of photoreceptor:

Feature	Rod Cells	Cone Cells
Light sensitivity	Very sensitive — work in dim light	Less sensitive — only work in bright light
Visual pigment	Rhodopsin (bleached by low light intensity)	Three types of iodopsin (red, green, blue sensitive)
Colour vision	No — only produce black/white images	Yes — different cone types detect different wavelengths
Distribution	Mostly peripheral retina	Concentrated at fovea (centre of retina)
Visual acuity	Low — many rods converge onto one bipolar cell (retinal convergence/summation)	High — usually 1:1 ratio with bipolar cells at fovea
Sensitivity vs acuity	High sensitivity, low acuity	Low sensitivity, high acuity

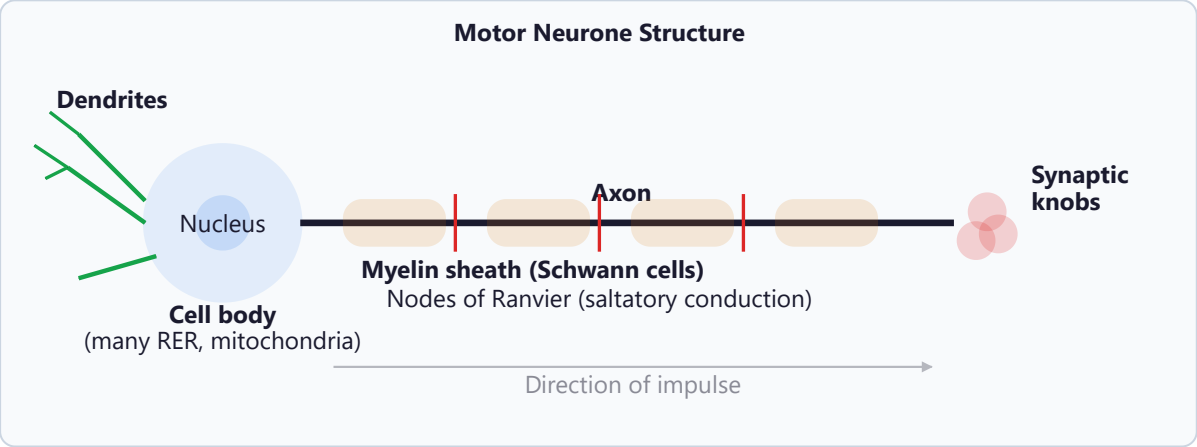
Retinal convergence: Many rod cells synapse with a single bipolar neurone. This means the individual generator potentials from multiple rods are summed (**spatial summation**), making rods very sensitive in dim light — but the brain cannot distinguish which rod was stimulated, so acuity is low. Cones have less convergence (especially at the fovea), giving higher acuity but requiring more light to reach threshold.

6.2 Nervous Coordination

Nervous system: Provides rapid, short-lived, specific responses using electrical impulses along neurones and chemical transmission at synapses. The central nervous system (CNS) consists of the brain and spinal cord. The peripheral nervous system (PNS) consists of sensory and motor neurones.



Neurone Structure



Types of Neurone

Type	Function	Structure
Sensory	Carries impulses from receptor to CNS	Long dendron, cell body off to side (in dorsal root ganglion), short axon into spinal cord
Motor	Carries impulses from CNS to effector	Short dendrites, cell body in CNS, long myelinated axon to effector
Relay (interneurone)	Connects sensory to motor neurones within CNS	Short, many connections; allows integration

The Nerve Impulse — Action Potential

Resting potential: The potential difference across a neurone membrane at rest, approximately **-70 mV** (inside negative relative to outside). Maintained by the **Na^+/K^+ ATPase pump** (actively transports 3 Na^+ out for every 2 K^+ in, using ATP) and by **K^+ leak channels** (K^+ diffuses out down its concentration gradient, making inside more negative). The membrane is said to be **polarised**.

Stages of an Action Potential

1. **Resting state (-70 mV):** Voltage-gated Na^+ and K^+ channels are closed. The Na^+/K^+ pump maintains the resting potential. The membrane is polarised.
2. **Stimulus received:** A stimulus (e.g. generator potential from a receptor, or depolarisation from adjacent region) causes some voltage-gated Na^+ channels to open.
3. **Depolarisation:** If the threshold potential (~-55 mV) is reached, many more voltage-gated Na^+ channels open (**positive feedback**). Na^+ rushes into the axon (down its electrochemical gradient). Membrane potential rises rapidly from -70 mV to about **+40 mV**. This is the **all-or-nothing** principle — if threshold is reached, a full action potential fires; if not, nothing happens.
4. **Repolarisation:** At +40 mV, voltage-gated Na^+ channels close (inactivation gates shut). Voltage-gated K^+ channels open (slightly delayed). K^+ flows out of the axon (down its concentration gradient), restoring the negative potential inside.
5. **Hyperpolarisation (undershoot):** K^+ channels are slow to close, causing a slight overshoot below -70 mV (to about -80 mV).
6. **Recovery:** K^+ channels finally close. The Na^+/K^+ pump restores the original ion distribution. The resting potential is restored.

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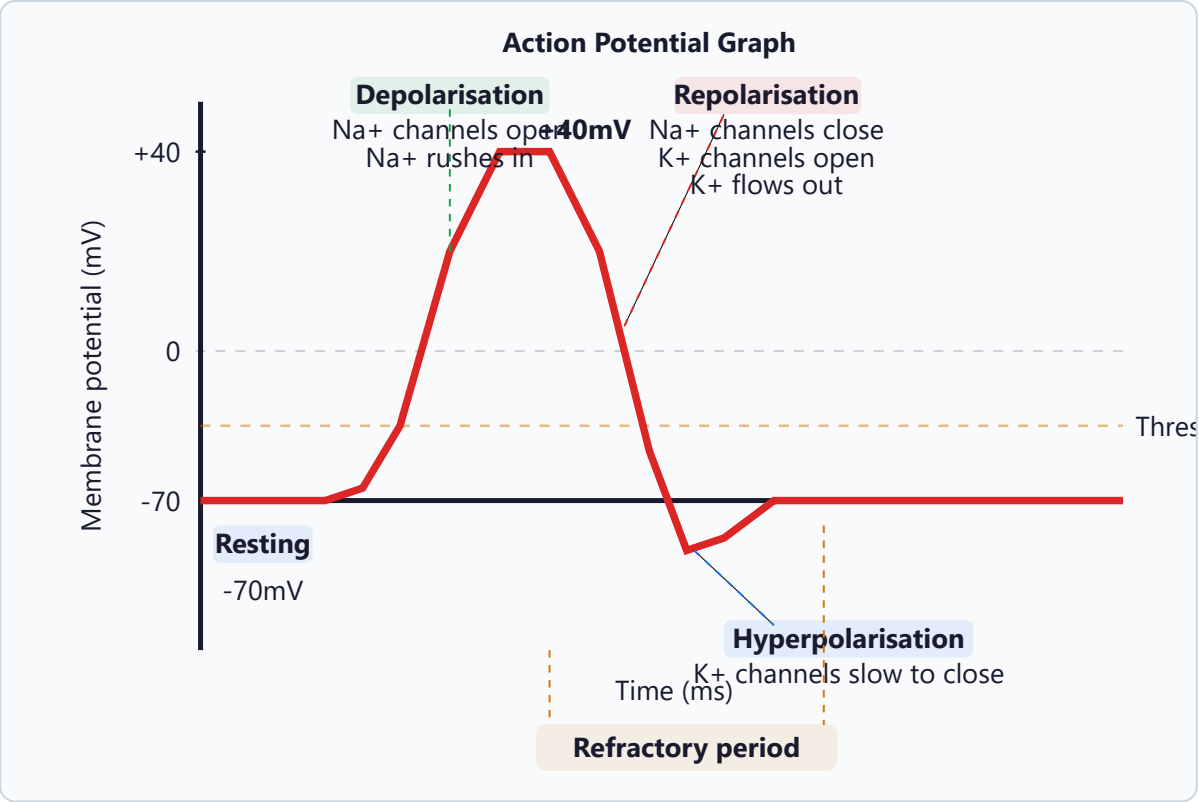


The Refractory Period

After an action potential, Na^+ channels are temporarily inactivated and cannot be reopened immediately. This is the refractory period, and it has three crucial functions:

- **Ensures unidirectional propagation:** The region that has just fired cannot be re-stimulated, so the impulse can only travel forward along the axon.
- **Limits impulse frequency:** Sets a maximum firing rate (~500-1000 impulses per second).
- **Ensures discrete impulses:** Each action potential is a separate event; impulses do not merge together.

All-or-Nothing Principle

A nerve impulse either fires completely (+40 mV) or not at all. There is no such thing as a "bigger" action potential. So how does the nervous system distinguish between strong and weak stimuli?

- **Frequency coding:** A stronger stimulus produces more action potentials per second (higher frequency), not bigger ones.
- **Recruitment:** A stronger stimulus activates more neurones simultaneously.

Transmission Speed

Factors affecting the speed of impulse transmission:

Factor	Effect	Explanation
Myelination	Greatly increases speed (up to 120 m/s vs 0.5-2 m/s unmyelinated)	Myelin sheath (formed by Schwann cells) insulates the axon. Depolarisation only occurs at **nodes of Ranvier** (gaps ~1 mm apart). The impulse "jumps" between nodes — **saltatory conduction** — much faster than continuous conduction.
Axon diameter	Wider = faster	Wider axons have less electrical resistance to local current flow. Giant squid axon (~1 mm diameter) is an example of an unmyelinated but fast axon.
Temperature	Higher = faster (up to a point)	Increases rate of ion diffusion and enzyme activity (Na^+/K^+ ATPase). Above ~40°C, proteins denature.

Synapses

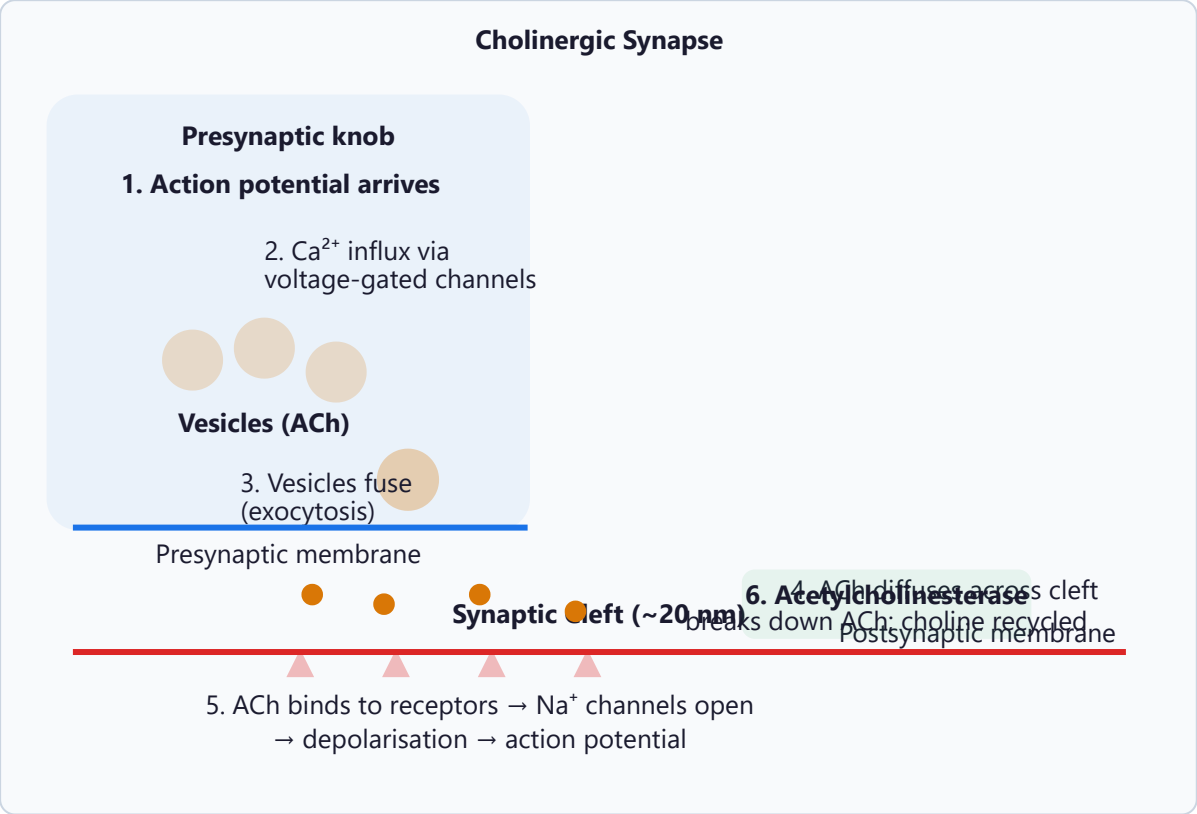
Synapse: The junction between two neurones (or between a neurone and an effector). There is a **synaptic cleft** (~20 nm gap) between them. Transmission is typically chemical, using **neurotransmitter molecules** that diffuse across the cleft. Synapses are crucial for neural processing, integration, and decision-making.

Cholinergic Synaptic Transmission

1. **Action potential arrives** at the presynaptic knob (axon terminal).
2. **Voltage-gated Ca^{2+} channels open** in the presynaptic membrane. Ca^{2+} ions flood into the presynaptic knob (down their concentration gradient).
3. Ca^{2+} ions cause **synaptic vesicles** (containing acetylcholine, ACh) to move towards and **fuse with the presynaptic membrane** (exocytosis), releasing ACh into the synaptic cleft.
4. ACh **diffuses across the cleft** (takes ~0.5 ms) and binds to specific **nicotinic ACh receptors** on the postsynaptic membrane. These receptors are ligand-gated (chemically gated).
5. Binding of ACh causes **Na^+ channels** on the postsynaptic membrane to open. Na^+ influx causes **depolarisation** (an excitatory postsynaptic potential, EPSP). If this reaches threshold, an action

potential is generated in the postsynaptic neurone.

6. ACh is rapidly broken down by **acetylcholinesterase** (AChE) in the cleft into choline and ethanoic acid. Choline is reabsorbed into the presynaptic knob by active transport and recycled to make more ACh (using choline acetyltransferase and ATP).



Functions of Synapses

- **Unidirectional transmission:** Neurotransmitter receptors are only on the postsynaptic membrane; vesicles are only in the presynaptic knob. Impulses can only cross in one direction.
- **Summation:** Allows integration of signals from multiple sources:
  - **Temporal summation:** Rapid repeated impulses from the same presynaptic neurone build up neurotransmitter, reaching threshold in the postsynaptic neurone
  - **Spatial summation:** Simultaneous impulses from multiple presynaptic neurones converging on one postsynaptic neurone combine their effects to reach threshold
- **Inhibition:** Inhibitory neurotransmitters (e.g. GABA, glycine) cause Cl^- influx or K^+ efflux → **hyperpolarisation** (inhibitory postsynaptic potential, IPSP) → makes an action potential less likely
- **Integration:** The postsynaptic neurone "adds up" all EPSPs and IPSPs. If the net effect exceeds threshold, it fires; if not, it does not. This allows complex processing and decision-making.

- **Amplification:** One action potential can cause release of many vesicles of neurotransmitter, activating many ion channels — amplifying the signal.

6.3 Muscle Contraction — The Sliding Filament Theory

Sliding Filament Theory: Muscle contraction occurs when **actin (thin) filaments slide over myosin (thick) filaments**, shortening the sarcomere. The filaments themselves do not shorten — they slide past each other using **cross-bridge cycling** powered by ATP hydrolysis.

Structure of Skeletal Muscle

The hierarchical organisation: **Muscle organ** → **fascicles (bundles)** → **muscle fibres (cells)** → **myofibrils** → **sarcomeres** (the functional contractile units).

Key features of muscle fibres:

- **Multinucleate** — many nuclei per fibre (formed by fusion of myoblasts during development)
- **Sarcolemma** — the cell surface membrane of a muscle fibre
- **T-tubules** — infoldings of sarcolemma that carry action potentials deep into the fibre
- **Sarcoplasmic reticulum (SR)** — specialised smooth ER that stores and releases Ca^{2+} ions
- **Many mitochondria** — for aerobic ATP production
- **Myoglobin** — oxygen-storing protein (red pigment), similar to haemoglobin

Sarcomere Bands and Zones

Band/Zone	Contains	During Contraction
I band (light)	Actin only	Gets **shorter**
A band (dark)	Myosin (+ overlapping actin)	Stays the **same length**
H zone	Myosin only (centre of A band)	Gets **shorter**
Z line (Z disc)	Boundary of sarcomere; anchors actin	Z lines move **closer together**

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M line	Centre of sarcomere; anchors myosin	Stays in centre

Exam Tip: During contraction: I band shortens, H zone shortens, A band stays the same, sarcomere shortens. The key point: the **A band does not change length** because the myosin filaments do not get shorter — they remain the same length. Only the regions where there is no overlap (I band, H zone) get shorter as actin slides over myosin. This is the most commonly tested detail.

Mechanism of Contraction — Cross-Bridge Cycling

1. **Nerve impulse arrives** at the neuromuscular junction (NMJ). ACh is released, crossing the synaptic cleft and binding to receptors on the sarcolemma. This depolarises the muscle fibre membrane.
2. The impulse travels along the **sarcolemma** and down **T-tubules** deep into the fibre.
3. T-tubule depolarisation triggers the **sarcoplasmic reticulum (SR)** to release stored **Ca^{2+} ions** into the sarcoplasm (via calcium release channels).
4. Ca^{2+} binds to **troponin** (a protein on the actin filament). This causes a conformational change that moves **tropomyosin** away from the **myosin binding sites** on actin, exposing them.
5. **Cross-bridge formation:** The myosin head (which has ADP + P_i bound from previous ATP hydrolysis) attaches to the now-exposed binding site on actin, forming a cross-bridge (actomyosin complex).
6. **Power stroke:** ADP + P_i are released from the myosin head. This causes the head to pivot/tilt (~45°), pulling the actin filament towards the centre of the sarcomere. The sarcomere shortens.
7. **Detachment:** A new ATP molecule binds to the myosin head, causing it to detach from actin (breaking the cross-bridge).
8. **Recovery stroke:** ATP is hydrolysed to ADP + P_i (by myosin ATPase). This provides energy to re-cock the myosin head back to its high-energy position, ready to form a new cross-bridge further along the actin. The cycle repeats as long as Ca^{2+} and ATP are available.

Relaxation

When nervous stimulation stops:

- Ca^{2+} is actively pumped back into the SR by **Ca^{2+}-ATPase** (requires ATP)
- Ca^{2+} concentration in sarcoplasm drops; Ca^{2+} dissociates from troponin
- Tropomyosin re-covers the myosin binding sites on actin

- No more cross-bridges can form; muscle passively returns to resting length (stretched by antagonistic muscle contraction)

Exam Tip: ATP is needed for THREE processes during muscle activity: (1) cross-bridge cycling (ATP hydrolysis cocks the myosin head and allows detachment); (2) active transport of Ca^{2+} back into the SR (relaxation); (3) the Na^{+}/K^{+} pump to maintain resting potential. This explains **rigor mortis** — after death, ATP production stops. Without ATP, myosin heads cannot detach from actin, so muscles become locked in a rigid state. Rigor mortis develops 2-6 hours after death and persists until proteins degrade.

Slow-Twitch vs Fast-Twitch Muscle Fibres

Feature	Slow-Twitch (Type I)	Fast-Twitch (Type II)
Contraction speed	Slow, sustained	Fast, powerful, short bursts
Main energy source	Aerobic respiration	Anaerobic (glycolysis) + some aerobic
Mitochondria	Many	Fewer
Myoglobin	Lots (red colour)	Less (paler)
Glycogen stores	Less	More
Fatigue resistance	High (endurance)	Low (fatigue quickly)
Example muscles	Postural muscles (back, soleus)	Sprinting muscles (gastrocnemius)

6.4 Homeostasis

Homeostasis: The maintenance of a relatively constant internal environment within tolerable limits, despite changes in the external environment. It involves **receptors** detecting deviations from a set point, a **coordination centre** processing information, and **effectors** bringing about corrective responses. Most homeostatic mechanisms use **negative feedback**.

Negative Feedback

Negative feedback is the mechanism by which a deviation from a set point triggers a response that **returns the system to the set point** (opposes the original change). It maintains stability but cannot maintain a perfectly constant value — there is always a slight oscillation around the set point.

> **Positive feedback:** A deviation from the set point triggers a response that **amplifies the change further**. This is destabilising but useful in specific situations: action potential generation (Na^+ influx opens more Na^+ channels), blood clotting cascade, oxytocin during childbirth (contractions stimulate more oxytocin release), and fruit ripening (ethylene production stimulates more ethylene).

Blood Glucose Regulation

Blood glucose concentration is normally maintained at approximately **4-8 mmol/L** (or 80-120 mg/dL). It is regulated by two antagonistic hormones from the pancreas:

Condition	Detected by	Hormone	Target Cells	Effect
High blood glucose (after meal)	Beta (β) cells of islets of Langerhans (pancreas)	**Insulin**	Liver, muscle, adipose tissue	Stimulates **glycogenesis** (glucose → glycogen for storage); increases glucose uptake via **GLUT4 transporters** (insulin causes vesicles containing GLUT4 to fuse with cell membrane); stimulates lipogenesis (glucose → fat); stimulates cellular respiration of glucose
Low blood glucose (fasting/exercise)	Alpha (α) cells of islets of Langerhans	**Glucagon**	Liver cells (hepatocytes) mainly	Stimulates **glycogenolysis** (glycogen → glucose); stimulates **gluconeogenesis** (amino acids/glycerol → glucose); glucose released into blood

Second messenger model for glucagon:

1. Glucagon (first messenger) binds to a specific receptor on the hepatocyte cell surface membrane.

2. This activates the enzyme **adenylate cyclase** on the inner surface of the membrane (via a G-protein).
3. Adenylate cyclase converts ATP to **cyclic AMP (cAMP)** — the second messenger.
4. cAMP activates protein kinase A, which activates a phosphorylase enzyme cascade.
5. Phosphorylase catalyses glycogenolysis (glycogen → glucose-1-phosphate → glucose).
6. Glucose is released from the liver cell into the blood.

Adrenaline also raises blood glucose (using the same second messenger pathway via cAMP in liver cells). This is part of the "fight or flight" response.

Diabetes Mellitus

Feature	Type I	Type II
Cause	Autoimmune destruction of β cells → no insulin produced	Target cells become resistant to insulin (receptor downregulation) and/or reduced insulin secretion
Onset	Usually childhood/young adult	Usually later in life (but increasing in young people)
Risk factors	Genetic predisposition, possible viral trigger	Obesity, sedentary lifestyle, genetic predisposition, age, ethnicity
Treatment	Insulin injections (or pump), blood glucose monitoring, carbohydrate counting	Diet and exercise, metformin (increases cell sensitivity to insulin), eventually may need insulin
Prevalence	~10% of diabetes cases	~90% of diabetes cases

Body Temperature Regulation (Thermoregulation)

The **thermoregulatory centre** in the **hypothalamus** monitors core blood temperature directly (central thermoreceptors) and receives input from peripheral thermoreceptors in the skin.

Response to Overheating	Response to Cooling

Vasodilation of skin arterioles — more blood flows near skin surface — more heat lost by radiation, conduction, convection	**Vasoconstriction** — less blood near skin surface — less heat loss by radiation
Sweating increases — evaporation of water from skin surface removes latent heat of vaporisation (~2.4 kJ per gram)	**Shivering** — involuntary rhythmic muscle contractions generate heat from increased respiration
Hairs lie flat (erector pili muscles relax) — thin insulating air layer	**Hairs raised** (erector pili muscles contract via sympathetic nerves) — traps insulating layer of still air (less significant in humans)
Reduced metabolic rate; behavioural responses (seeking shade, removing clothing)	Increased metabolic rate — **adrenaline** and **thyroxine** increase cellular respiration rate; behavioural responses (curling up, adding clothing)

Exam Tip: Be precise about vasodilation/vasoconstriction. It is the **arterioles** supplying the skin capillary beds that dilate or constrict — not the capillaries themselves (capillaries have no muscle in their walls). When arterioles dilate, more blood flows through superficial capillary beds near the skin surface. When they constrict, blood is shunted through deeper shunt vessels, bypassing superficial capillaries.

The Kidney and Osmoregulation

The kidney maintains the water potential of blood by adjusting the volume and concentration of urine produced. Each kidney contains approximately 1 million **nephrons** (the functional units).

Nephron Function

- **Ultrafiltration (Bowman's capsule / glomerulus):** High hydrostatic pressure in the glomerulus (due to afferent arteriole being wider than efferent) forces small molecules (water, glucose, urea, amino acids, ions) through the basement membrane (acts as a molecular sieve) and through podocyte slits into the Bowman's capsule. Blood cells and large plasma proteins are too large to pass through and remain in the blood. The filtrate entering the capsule is called **glomerular filtrate**.
- **Selective reabsorption (proximal convoluted tubule, PCT):** ~85% of filtrate is reabsorbed here. Glucose and amino acids are reabsorbed by active transport and co-transport (Na^+/glucose co-

transporter on the apical membrane, Na^+/K^+ ATPase on the basolateral membrane creates the Na^+ gradient). Water follows by osmosis. The PCT has microvilli (brush border) for large surface area, many mitochondria for ATP, and a rich blood supply.

- **Loop of Henle:** Creates and maintains a **concentration gradient** (increasing NaCl concentration) in the renal medulla using the **countercurrent multiplier** mechanism. The descending limb is permeable to water (leaves by osmosis) but not to ions. The ascending limb actively pumps Na^+ and Cl^- out (thick ascending limb) and is impermeable to water. This creates a hypertonic medulla essential for concentrating urine.
- **Distal convoluted tubule (DCT):** Fine-tuning of ion balance (Na^+/K^+ exchange regulated by aldosterone); some water reabsorption.
- **Collecting duct:** Passes through the hypertonic medulla. Permeability to water is controlled by **ADH** (antidiuretic hormone). If permeable, water leaves by osmosis (concentrated urine); if impermeable, water stays in the tubule (dilute urine).

ADH and Water Reabsorption

When blood water potential is low (dehydration / high solute concentration):

1. **Osmoreceptors** in the hypothalamus detect the low water potential (they lose water by osmosis and shrink, which stimulates them).
2. The hypothalamus stimulates the **posterior pituitary** to release more **ADH** (vasopressin) into the blood.
3. ADH travels in the blood to the kidneys, where it binds to specific receptors on the basolateral membrane of collecting duct cells.
4. This triggers a second messenger cascade (cAMP) that causes **vesicles containing aquaporin-2 water channels** to fuse with the apical (luminal) membrane.
5. More aquaporins in the membrane = **more water reabsorbed by osmosis** (following the osmotic gradient created by the loop of Henle in the medulla).
6. Result: **small volume of concentrated urine**. Blood water potential returns to normal (negative feedback).

When blood water potential is high (overhydration): Less ADH released → fewer aquaporins → collecting duct less permeable to water → water stays in the tubule → **large volume of dilute urine**.

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Exam Tip: ADH makes the collecting duct MORE permeable to water — it does not "move water." Water moves by osmosis down the water potential gradient created by the countercurrent multiplier in the loop of Henle. ADH simply inserts aquaporin channels that allow this osmotic movement to occur. Without the medullary gradient, ADH alone would not concentrate urine. Also note: alcohol inhibits ADH secretion, which is why alcohol consumption leads to increased urine output (diuresis) and dehydration.

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